THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM THREE ANNUAL EXAMINATION
031 PHYSICS
MARKING SCHEME NOVEMBER 2025

SECTION A (16 MARKS)

1.

i.	ii.	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.
B	B	C	C	B	B	C	A	A	D

01 mark @ = 10 marks total

2.

i.	ii.	iii.	iv.	v.	vi.
F	E	B	A	D	C

01 mark @ = 06 marks total

SECTION B (54 MARKS)

3. (a) Plane mirror: Image is same size, upright, and virtual.02marks
Convex mirror: Image is smaller, upright, and virtual. 02marks

(b)If a convex lens forms a **diminished real image**, the object (candle) is **beyond 2F**…….. 03marks

Reason: A convex lens produces a smaller, real, and inverted image when the object is beyond twice the focal length. …….. 02marks

4. (a)Triangle law: When two vectors are represented in magnitude and direction by two sides of a triangle taken in order, their resultant is given by the third side. …….. 02marks
Parallelogram law: If two vectors act simultaneously at a point, the resultant is represented by the diagonal of the parallelogram formed by the vectors…….. 02marks
(b) ……. 05marks

Given: $F_1 = 25$ N, $F_2 = 35$ N, angle = 90°

$$R = \sqrt{F_1^2 + F_2^2} = \sqrt{25^2 + 35^2} = \sqrt{625 + 1225} = \sqrt{1850} = 43.0 \text{ N}$$

$$\tan\theta = \frac{F_2}{F_1} = \frac{35}{25} = 1.4 \implies \theta = 54.5^\circ$$

Note a vector diagram can also be used in calculation of the resultant vectors given in above question

5. (a) Efficiency = (Load × Height) / (Effort × Length) × 100……..02marks
= (500 × 0.6) / (120 × 3) × 100……..01marks
= (300) / (360) × 100……..01marks
= **83.3%**……..01marks
(b) Change in velocity = (–12) – 18
= –30 m/s
Change in momentum = 0.15 × (–30)
= –4.5 Ns……..01marks
Average force = (Δp) / t ……..01marks
= (4.5 / 0.1) ……..01marks
= 45 N (opposite to direction……..01marks

6. (a) Pressure difference = 76 – 70 = 6 cm Hg
Using ρ = 13600 kg/m³, g = 10 m/s²,
1 cm Hg = 1330 Pa……..01marks

ΔP = 6 × 1330 = 7980 Pa……..01marks

Height = ΔP / (ρair × g) ……..01marks

= 7980 / (1.25 × 10)
= 638.4 m……..01marks
(b) Let reactions at supports A and B be R_1 and R_2
Σ Moments about A = 0……..01marks
$R_2 \times 4 = (200 \times 1) + (300 \times 2)$ ……..01marks
$R_2 \times 4 = 200 + 600 = 800 \rightarrow R_2 = 200$ N……..01marks
Total load = 500 N……..01marks
$R_1 = 500 – 200 = 300$ N……..01marks

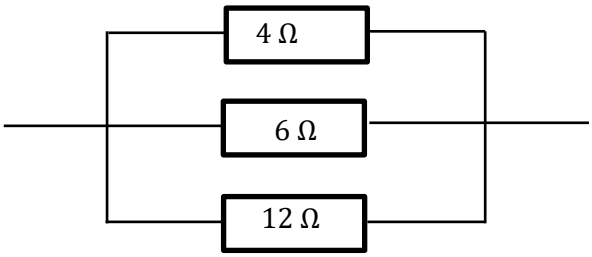
7. (a) The clay sticks to the floor → Perfectly inelastic collision……..03marks
Momentum: Conserved. ……..03marks
Kinetic energy: Not conserved (part converted to heat and deformation energy).
……..03marks
(b) this concept is not taught in o -level , so the marks are distributed at the top Iin 7(a)

8. (a) Uses of solar energy:
Drying clothes and grains. ……..01marks
Heating water (solar heaters). ……..01marks
Generating electricity (solar panels). ……..01marks
Cooking (solar cooker). ……..01marks
(b)Diagram (description):
The Sun at the center.

Plants (photosynthesis) → source of biomass.
Wind, rainfall, and water energy derived from the Sun.
Solar panels directly convert sunlight into electricity.……..05marks

SECTION C (30 MARKS)

9. (a) (i) three resistors are connected in parallel between two point 2.5Mrks



(ii) $from\ \frac{1}{Rt} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12}$…….....01mark

Rt= 2Ω…….....01.5mark

Total capacitance $C_T = C_1 + C_2 + C_3 = 5 + 10 + 15$…….....01mark

$= 30\mu F$…….....02mark
Advantage: If one capacitor fails, others still function (increases total capacity).…….....02mark

(c)(i) Lightning: A sudden electrostatic discharge between clouds and Earth.....01mark
(ii) Protection:
Install lightning conductors.
Use earthing systems.
Avoid tall metal structures.
Disconnect electrical devices during storms.…….....01mark@= 04 marks

10. (a)
Additive mixing:

Combining red, green, blue light → forms white
…….....01mark
used in TV, screens).…….....01.5mark
Subtractive mixing**:**
Pigments absorb colors; mixing cyan, magenta, yellow gives
black…….....01mark
used in printing…….....01mark
(b) Devices using total internal reflection:
Optical fibers.…….....01.5mark
Periscopes.…….....01.5mark

(c) Given h_0 = 2cm , u= 12cm , f=8cm
Required image distance, magnification and nature of the image formed
Using the formulae
$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$…….....01mark
making v thee subject of the formulae we have $v = \frac{fu}{u-f}$
$= \frac{8cm \times 12cm}{12cm - 8cm}$…….....01mark
$= 24cm$…….....01mark
$\therefore$ $image\ distance =$
$24cm\ from\ the\ mirror$ …….....01/2mark

- Magnification

Magnification $= \frac{24cm}{12cm}$…….....01/2mark

= 2

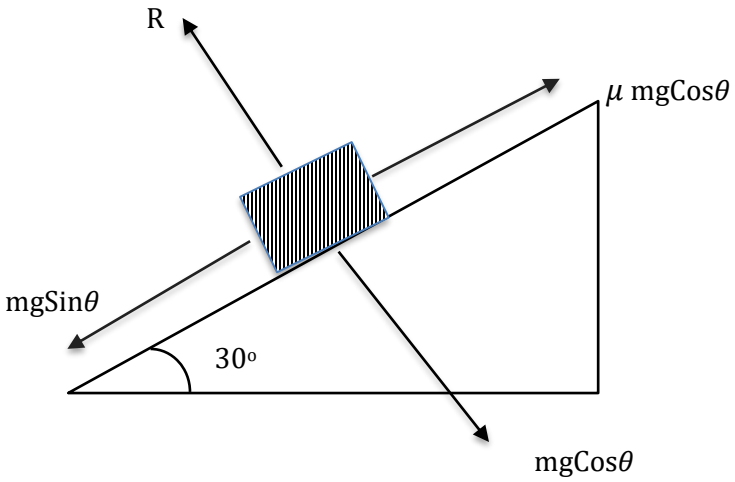
- Nature of image formed

The image formed
Real , inverted, magnified

11. (a) The heavily loaded bicycles is more worn out because it experienced larger friction force , the more the mass the large the friction force from $f = \mu mg$ friction force depends on mass i.e. friction force is directly proportion to mass this makes the heavily loaded bicycle experience more friction force than the light one…….03marks
(b) the coefficient of friction is a number without units because it is given as a ration between two physical quantities of the same SI units i.e. both frictional force and the normal reaction has the same units of Newtons ($\mu = \frac{friction\ force\ N}{Normal\ reaction\ N}$……..2marks

(c) given mass M= 3kg , $\theta = 30^{\circ}$, accelerationa=$3m/s^2$, required coefficient of friction μ

Consider the diagram bellow ……01marks



From the diagram above Net force =$mgsin\theta - \mu mgcos\theta$……01marks

But net force = ma where a represents acceleration and m represents mass of given body

Then $ma = mgsin\theta - \mu mgcos\theta$ make μ the subject of the formulae we have $\mu = \frac{mgsin\theta - ma}{mgcos\theta} \rightarrow \mu = \frac{gsin\theta - a}{gcos\theta}$

Substituting the given value we $\mu = \frac{10m/s^2 sin30^\circ - 4m/s^2}{10m/s^2 cos30^\circ}$……01marks

On simplifying =0.12 ……01marks

Therefore, the coefficient of friction is 0.12 ……01marks

(d) (i) three activities which could be impossible if there was no friction on the earth are :

- Walking on the ground could be impossible or difficult
- Car would not move because for the car to move there are must be friction between the car tyre and the ground
- There are could be no stopping or slowing down of the moving objects
- Sharpening of the tools could not be easy without friction
- Lighting matches could not be possible
- Setting off fire by rubbing sticks could be impossible
- Fixing nails in woods or walls could not be possible if there is no friction

- Writing on the paper could impossible any three activities 3marks

  (ii) benefits of friction are
- Friction aids in walking
- Makes cars to move
- Aids the moving objects to slow down
- Aids in lighting matches any two benefits 02marks 1@